

DSphinx: A Decentralized Path Selection for Decentralized Networks

at CBT Workshop (ESORICS)

by Aurélien Chassagne

on September 17, 2026



Problem Statement

Decentralized Route Selection

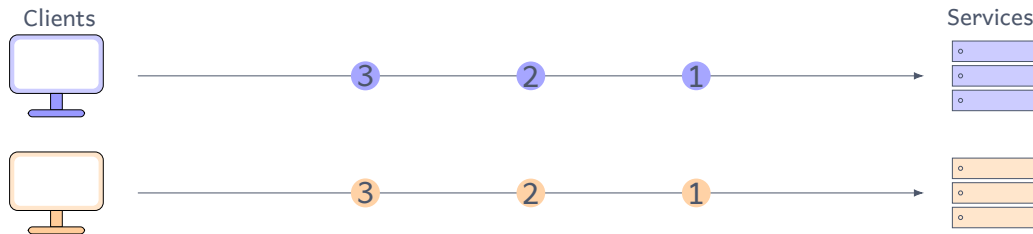
Conclusion

Problem Statement

Decentralized Route Selection

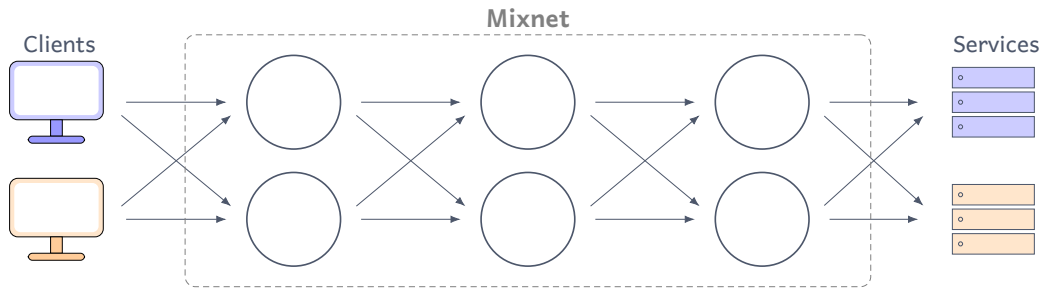
Conclusion

» Introduction



Despite encryption, metadata reveals who is communicating to whom

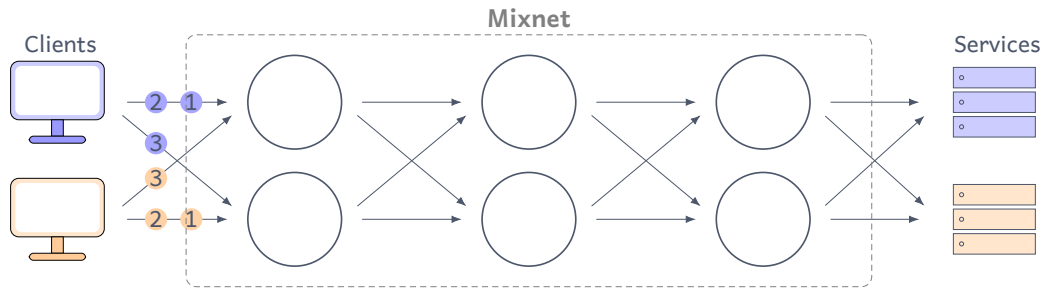
» Mixnet: A Privacy-Preserving Network



Unlinkability {

- * Fixed packet size
- * Shuffling packet

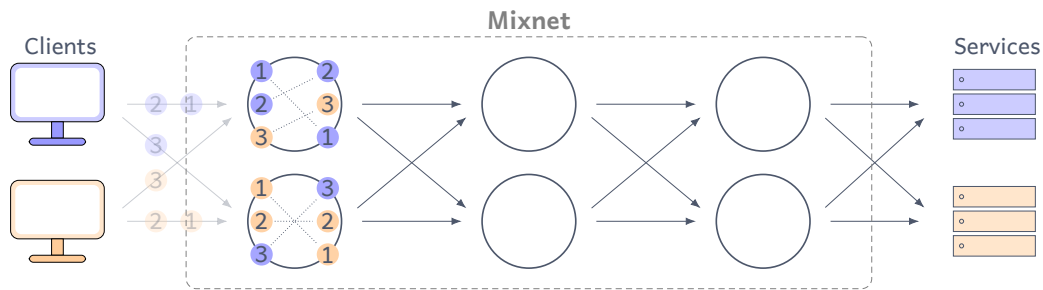
» Mixnet: A Privacy-Preserving Network



Unlinkability {

- * Fixed packet size
- * Shuffling packet

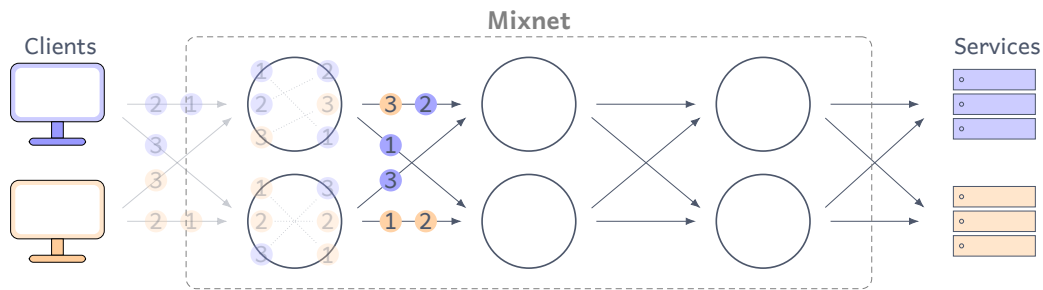
» Mixnet: A Privacy-Preserving Network



Unlinkability {

- * Fixed packet size
- * Shuffling packet

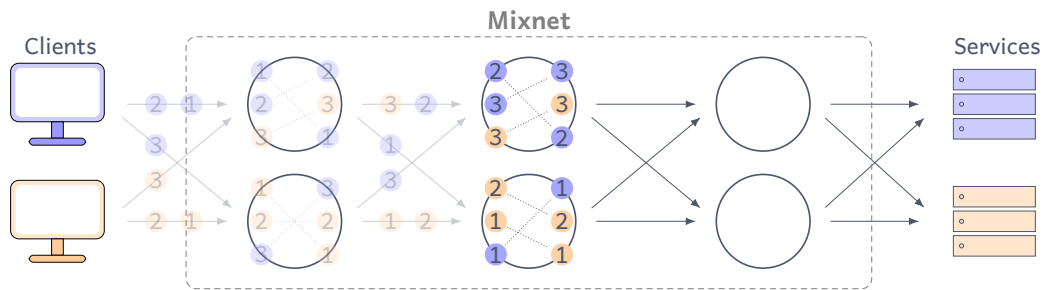
» Mixnet: A Privacy-Preserving Network



Unlinkability {

- * Fixed packet size
- * Shuffling packet

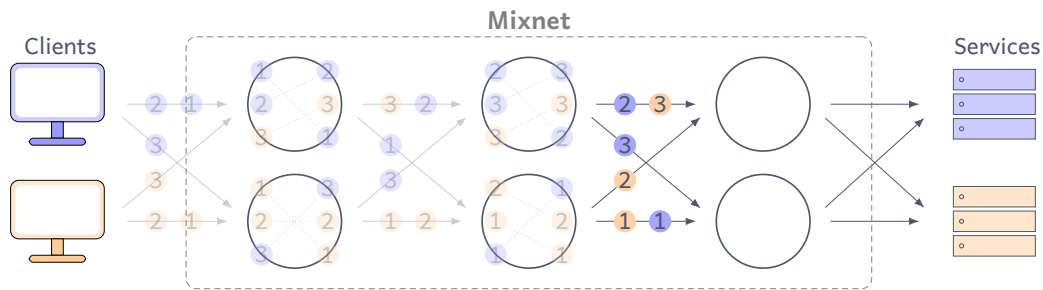
» Mixnet: A Privacy-Preserving Network



Unlinkability {

- * Fixed packet size
- * Shuffling packet

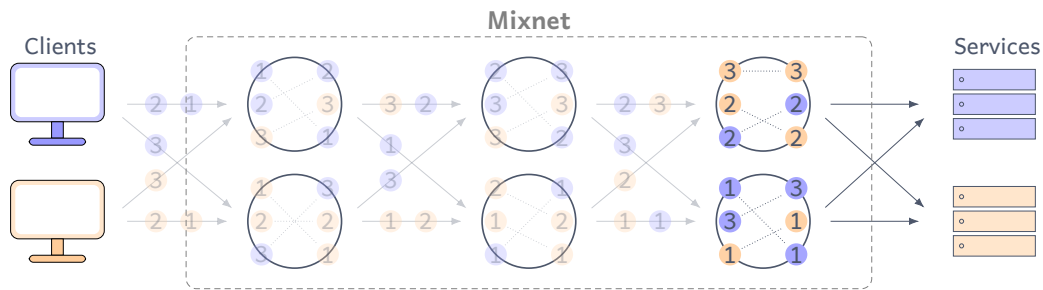
» Mixnet: A Privacy-Preserving Network



Unlinkability {

- * Fixed packet size
- * Shuffling packet

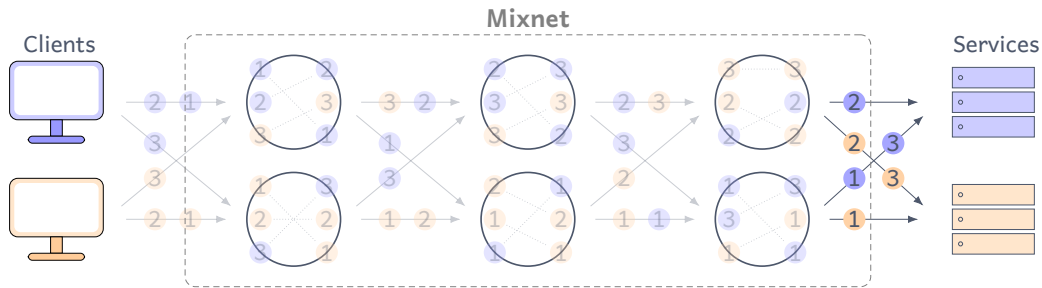
» Mixnet: A Privacy-Preserving Network



Unlinkability {

- * Fixed packet size
- * Shuffling packet

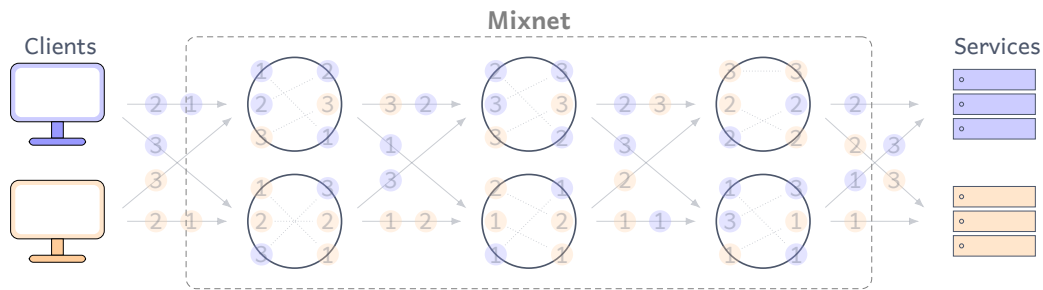
» Mixnet: A Privacy-Preserving Network



Unlinkability {

- * Fixed packet size
- * Shuffling packet

» Mixnet: A Privacy-Preserving Network



Unlinkability {

- * Fixed packet size
- * Shuffling packet

» The Client Controls Route Selection

Client
chooses route

» The Client Controls Route Selection

Malicious Client
chooses route



**Route-based
Attacks**

» The Client Controls Route Selection

Malicious Client
chooses route



Route-based Attacks

Anonymity-set reduction
 Packet tracing
 Targeting specific mixnodes

($N - 1$ attack)
 (DoS, reputation attack)

» The Client Controls Route Selection

Malicious Client
chooses route



Route-based Attacks

Anonymity-set reduction
 Packet tracing
 Targeting specific mixnodes (DoS, reputation attack) ($N - 1$ attack)

Research Question

How to prevent route manipulation without compromising privacy?

» The Client Controls Route Selection

Malicious Client
chooses route



**Route-based
Attacks**

Anonymity-set reduction
Packet tracing
Targeting specific mixnodes

($N - 1$ attack)
(DoS, reputation attack)

Research Question

How to prevent route manipulation without compromising privacy?

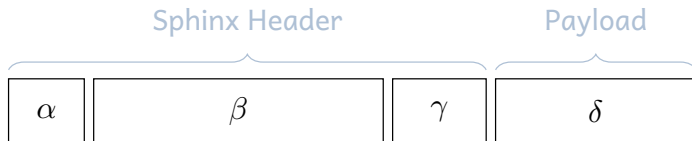
Our approach: Decentralize route sampling across STTPs

Problem Statement

Decentralized Route Selection

Conclusion

» Sphinx Packet



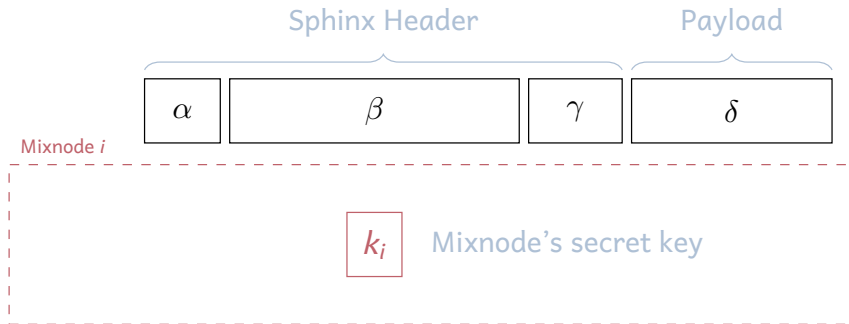
α : Cryptographic element

β : Encrypted routing information

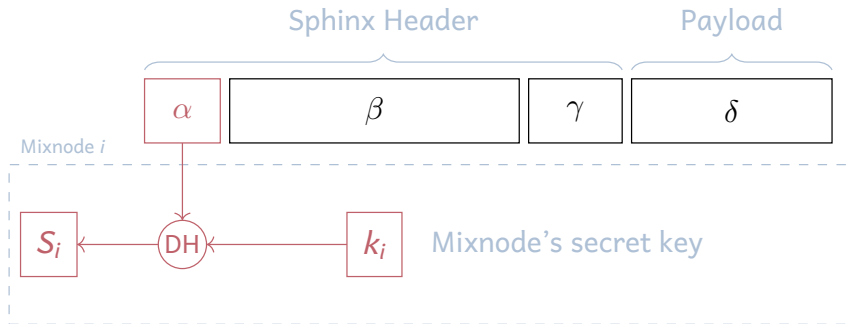
γ : Integrity check

δ : Payload

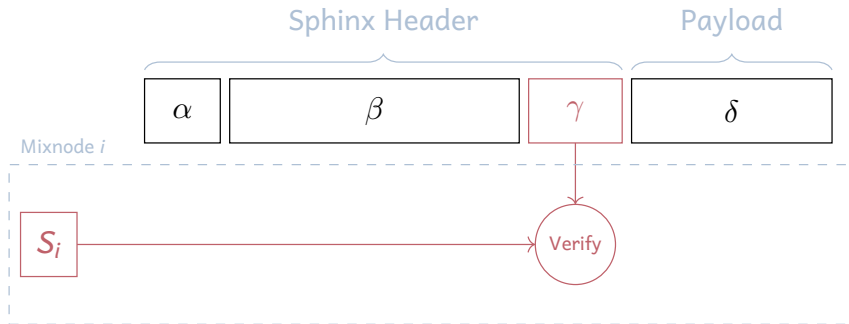
» Sphinx Packet



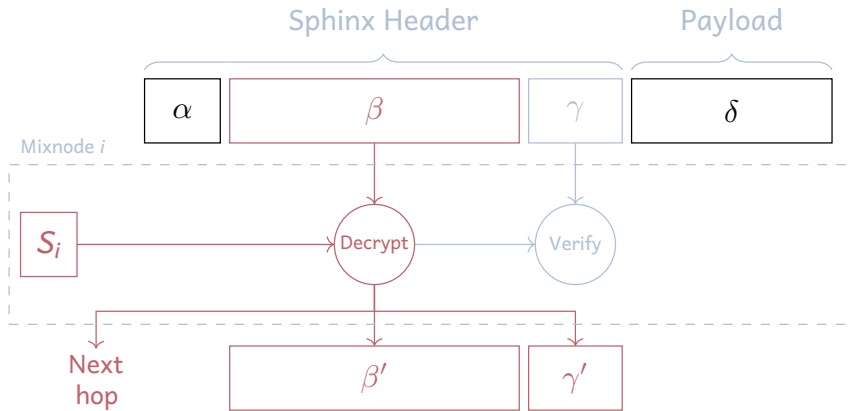
» Sphinx Packet



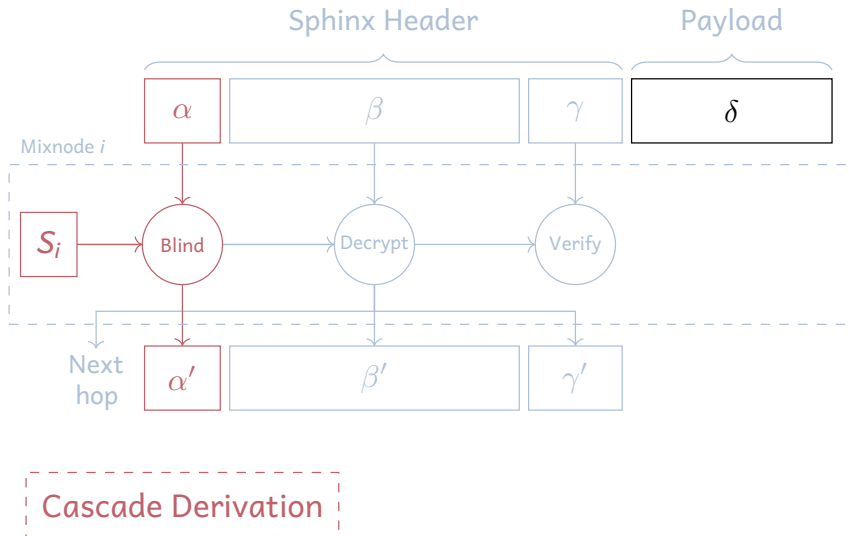
» Sphinx Packet



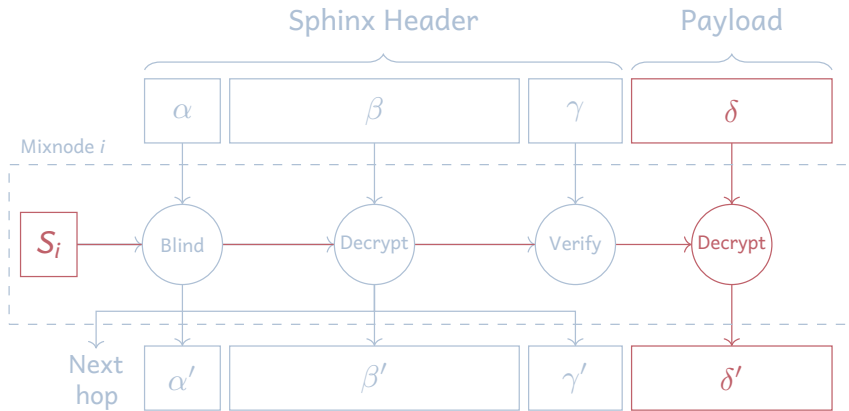
» Sphinx Packet



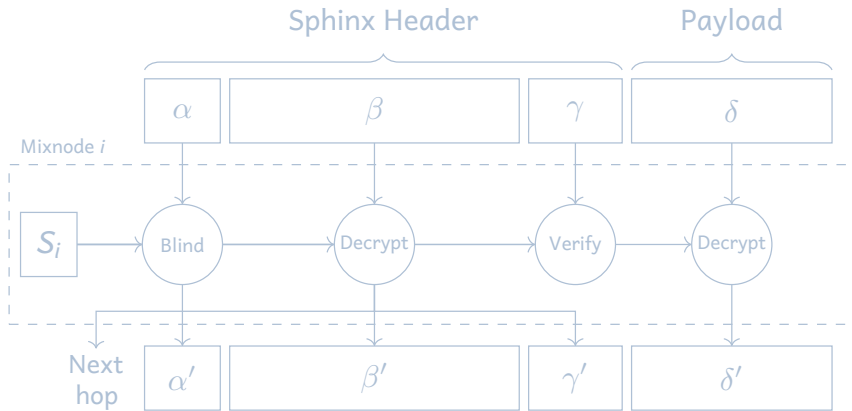
» Sphinx Packet



» Sphinx Packet



» Sphinx Packet



Properties

- * Per-hop Confidentiality
- * Per-hop Integrity
- * Fixed-Size

» Overview

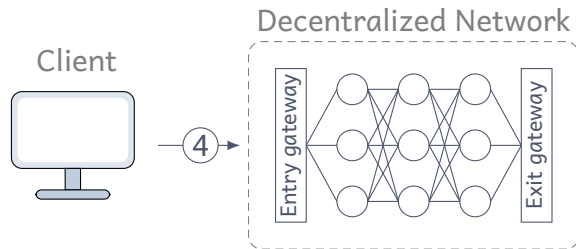


Figure: Overview of decentralized route sampling architecture.

» Overview

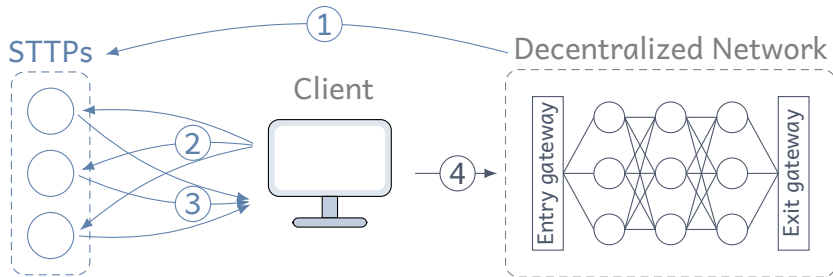
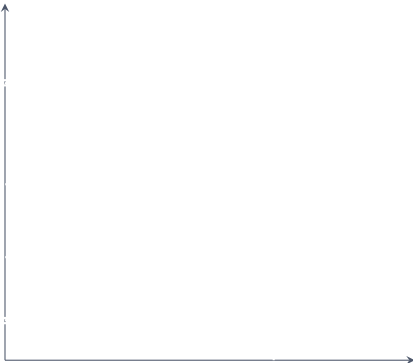


Figure: Overview of decentralized route sampling architecture.

- ① **Setup** stage (once)
- ② **Route sampling** request
- ③ **Aggregation** of sampled routes

» Setup Procedure

Mixnode i
Shamir Secret Sharing



» Setup Procedure

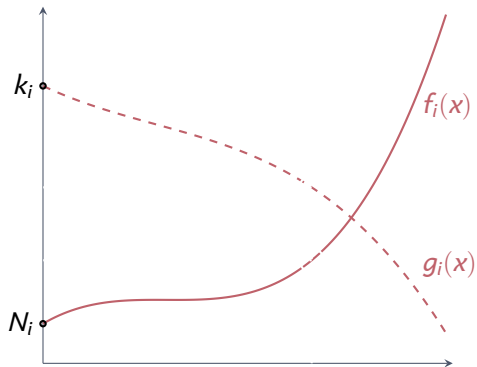
Mixnode i Shamir Secret Sharing



Mixnode i { random identifier r_i
secret key k_i
encoded address N_i

» Setup Procedure

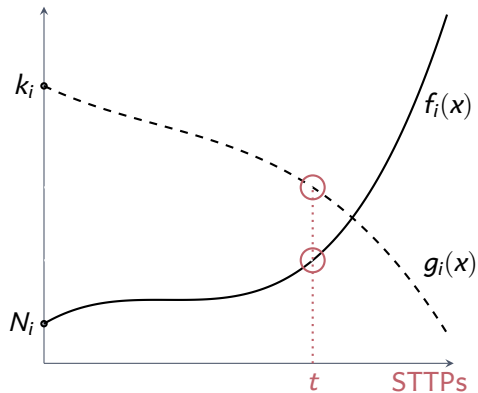
Mixnode i Shamir Secret Sharing



Mixnode i { random identifier r_i
secret key k_i
encoded address N_i

» Setup Procedure

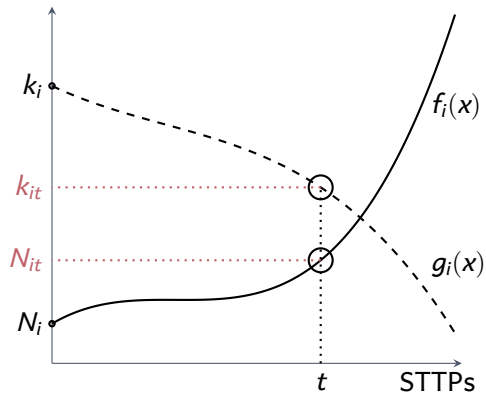
Mixnode i Shamir Secret Sharing



Mixnode i { random identifier r_i
secret key k_i
encoded address N_i

» Setup Procedure

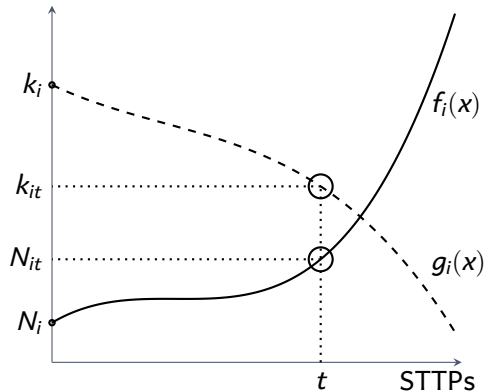
Mixnode i Shamir Secret Sharing



Mixnode i { random identifier r_i
secret key k_i
encoded address N_i

» Setup Procedure

Mixnode i Shamir Secret Sharing



Mixnode i { random identifier r_i
secret key k_i
encoded address N_i

STTP t Storing shares

idx	$f_i(t)$	$g_i(t)$
r_1	k_{1t}	N_{1t}
$\vdots$	$\vdots$	$\vdots$
r_i	k_{it}	N_{it}
$\vdots$	$\vdots$	$\vdots$
r_m	k_{mt}	N_{mt}

(r_i, k_{it}, N_{it})

» Route Sampling

STTP t *Partial Route Sampling*

Input: w

Client nonce



» Route Sampling

STTP t *Partial Route Sampling*

Input: w

Client nonce

$\ell \leftarrow \text{routeLength}$

$\alpha \leftarrow \text{Hash}(w)$



» Route Sampling

STTP t *Partial Route Sampling*

Input: w

Client nonce

$\ell \leftarrow \text{routeLength}$

$\alpha \leftarrow \text{Hash}(w)$

for $i = 1$ **to** ℓ **do**

end for



» Route Sampling

STTP t *Partial Route Sampling*

Input: w

Client nonce

$\ell \leftarrow \text{routeLength}$

$\alpha \leftarrow \text{Hash}(w)$

for $i = 1$ **to** ℓ **do**

$r'_i \leftarrow \text{Hash}(w, \text{timestamp}, i)$

Pseudorandom seed

end for



» Route Sampling

STTP t *Partial Route Sampling*

Input: w

Client nonce

$\ell \leftarrow \text{routeLength}$

$\alpha \leftarrow \text{Hash}(w)$

for $i = 1$ **to** ℓ **do**

$r'_i \leftarrow \text{Hash}(w, \text{timestamp}, i)$

Pseudorandom seed

$(k_{it}, N_{it}) \leftarrow \text{ClosestRow}(r'_i)$

Deterministic lookup

end for



» Route Sampling

STTP t *Partial Route Sampling*

Input: w

Client nonce

$\ell \leftarrow \text{routeLength}$

$\alpha \leftarrow \text{Hash}(w)$

for $i = 1$ **to** ℓ **do**

$r'_i \leftarrow \text{Hash}(w, \text{timestamp}, i)$

$(k_{it}, N_{it}) \leftarrow \text{ClosestRow}(r'_i)$

$S_{it} \leftarrow k_{it} \cdot \alpha$

end for

Pseudorandom seed

Deterministic lookup

Partial shared secret



» Route Sampling

STTP t *Partial Route Sampling*

Input: w

Client nonce

$\ell \leftarrow \text{routeLength}$

$\alpha \leftarrow \text{Hash}(w)$

for $i = 1$ **to** ℓ **do**

$r'_i \leftarrow \text{Hash}(w, \text{timestamp}, i)$

$(k_{it}, N_{it}) \leftarrow \text{ClosestRow}(r'_i)$

$S_{it} \leftarrow k_{it} \cdot \alpha$

Pseudorandom seed

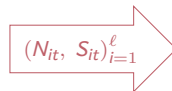
Deterministic lookup

Partial shared secret

end for

Output: $(N_{it}, S_{it})_{i=1}^{\ell}$

Partial route information



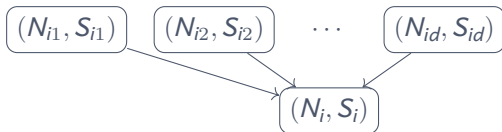
» Aggregation

Lagrange Polynomial Interpolation

$$N_i = \sum_{t=1}^d \lambda_t N_{it}$$

$$S_i = \sum_{t=1}^d \lambda_t S_{it}$$

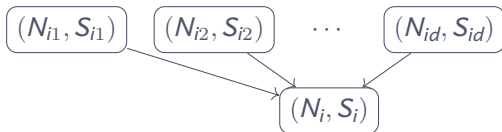
$\forall i \in \{1, \dots, \ell\}$



» Aggregation

Lagrange Polynomial Interpolation

$$N_i = \sum_{t=1}^d \lambda_t N_{it} \quad \forall i \in \{1, \dots, \ell\}$$
$$S_i = \sum_{t=1}^d \lambda_t S_{it}$$

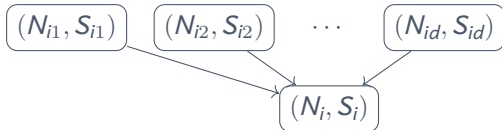


Shared secrets S_i are independent

» Aggregation

Lagrange Polynomial Interpolation

$$N_i = \sum_{t=1}^d \lambda_t N_{it} \quad \forall i \in \{1, \dots, \ell\}$$
$$S_i = \sum_{t=1}^d \lambda_t S_{it}$$



Shared secrets S_i are independent

Cascade Derivation (in Sphinx)

$$S_1 \longrightarrow S_2 \longrightarrow \dots \longrightarrow S_\ell$$

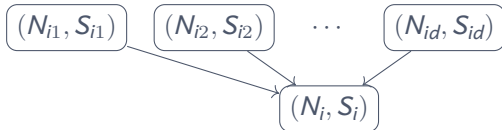
$$s_i = \text{Hash} \left(r S_i \prod_{j=1}^{i-1} s_j \right)$$

» Aggregation

Lagrange Polynomial Interpolation

$$N_i = \sum_{t=1}^d \lambda_t N_{it} \quad \forall i \in \{1, \dots, \ell\}$$

$$S_i = \sum_{t=1}^d \lambda_t S_{it}$$

Shared secrets S_i are independent

Cascade Derivation (in Sphinx)

$$S_1 \longrightarrow S_2 \longrightarrow \dots \longrightarrow S_\ell$$

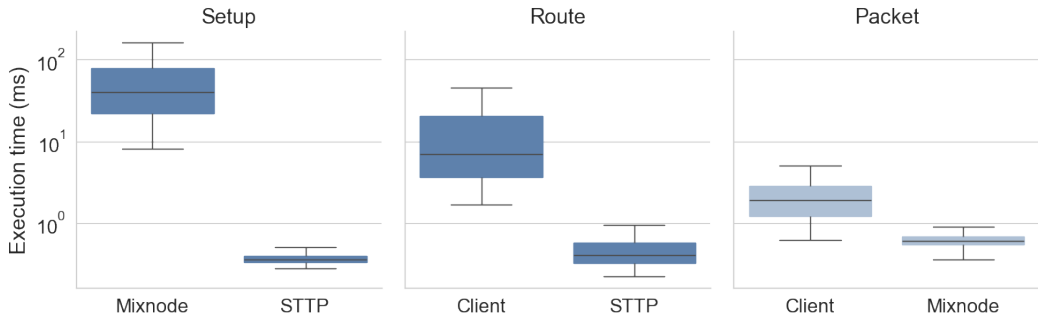
$$s_i = \text{Hash} \left(r S_i \prod_{j=1}^{i-1} s_j \right)$$

Where r is a nonce to rerandomize α

$$\alpha' = r \alpha$$

Fresh nonce $r \Rightarrow$ unlinkability

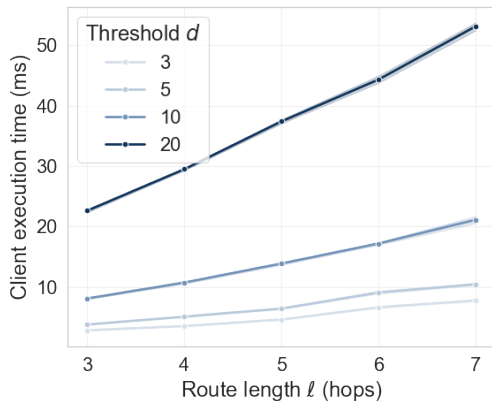
» Computation Time Across Protocol Stages



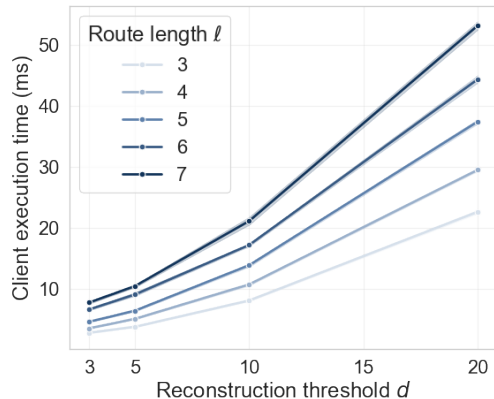
Overhead { Shamir secret sharing (Mixnode Setup)
Lagrange interpolation (Client Aggregation)

» Scalability with Respect to System Parameters

Route length



STTP threshold



Overhead remains moderate (50 ms) vs. end-to-end mixnet latency (500-800 ms)

Problem Statement

Decentralized Route Selection

Conclusion

» Future Work

Alternative approaches

ZK /VRF-based routing

Generalized route selection

Cost-based routing

Broader applications

Blockchain networks

» Takeaway

Prevent route manipulation

without compromising route privacy

Malicious client
Route bias



Decentralized sampling
Random route selection

Moderate overhead with preserved route confidentiality

DSphinx: A Decentralized Path Selection for Decentralized Networks

at CBT Workshop (ESORICS)

by Aurélien Chassagne

on September 17, 2026

